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VERTICAL RESISTIVE MEMORY DEVICE AND RELATED METHOD

Abstract

A device includes: a device layer; and an interconnect structure on the device layer, the interconnect structure including a resistive random access memory (RRAM) device. The RRAM device includes: an electrode layer above the device layer; an oxide semiconductor layer on the electrode layer; a gate structure that wraps around the oxide semiconductor layer; an insulating layer on the gate structure; and a resistor. The resistor includes: a bottom electrode on the oxide semiconductor layer; a dielectric layer on the bottom electrode; and a top electrode on the dielectric layer.

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Background/Summary

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIGS. 1A-1R are views of various embodiments of an IC device at various stages of fabrication according to aspects of the present disclosure.

[0004] FIGS. 2A and 2B are diagrammatic cross-sectional and circuit diagram views of various embodiments of an IC device according to aspects of the present disclosure.

[0005] FIGS. 3A-3U are views of various embodiments of an IC device at various stages of fabrication according to aspects of the present disclosure.

[0006] FIGS. 4A and 4B are diagrammatic cross-sectional and circuit diagram views of various embodiments of an IC device according to aspects of the present disclosure.

[0007] FIGS. 5 and 6 are flowcharts of methods of forming an IC device in accordance with various embodiments.

DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] Terms indicative of relative degree, such as “about,” “substantially,” and the like, should be interpreted as one having ordinary skill in the art would in view of current technological norms.

[0011] The terms “first,” “second,” “third” and so on may be used herein to describe a sequence of

events or sequential order of elements but may be exchanged or varied in some contexts. For example, a second layer may be formed on (e.g., sequentially after) a first layer, but in some contexts the first layer may be referred to as a “second layer,” “third layer,” “fourth layer” or the like, and the second layer may be referred to as a “first layer,” “third layer,” “fourth layer,” or the like.

[0012] The term “surrounds” may be used herein to describe a structure that fully or partially encloses another element or structure, for example, in three dimensions. For example, a first structure may “surround” a second structure on four lateral sides (e.g., left, right, front and back) without surrounding the second structure on two vertical sides (e.g., top and bottom). In other example, the first structure may wrap partially around the second structure, for example, by wrapping around three sides (e.g., top, front and back) while leaving other sides (e.g., left, right and bottom) exposed.

[0013] “Source/drain region(s)” may refer to a source or a drain, individually or collectively dependent upon the context.

[0014] The present disclosure is generally related to semiconductor devices, and more particularly to field-effect transistors (FETs), such as planar FETs, three-dimensional fin FETs (FinFETs), or nanostructure FETs, such as nanosheet FETs (NSFETs), nanowire FETs (NWFETs), gate-all-around FETs (GAAFETs) and the like.

[0015] Integrated memory refers to memory technologies that are built directly onto a microchip or integrated circuit, rather than being separate or “discrete” components. One such technology is Resistive Random Access Memory (RRAM), also known as ReRAM (Resistive RAM) or memristor-based memory. RRAM is a non-volatile memory technology that has benefits including high density, low power, and fast access.

[0016] RRAM operates on a principle of resistive switching. RRAM uses materials that can change their resistance state between a high-resistance (OFF) state and a low-resistance (ON) state in response to an applied voltage. These materials typically have a thin insulating layer sandwiched between two electrodes. By applying voltage pulses of selected magnitudes, the resistance of the insulating layer can be switched between its different states. RRAM is a non-volatile memory technology, which means it retains stored data even when power is turned off. RRAM devices use relatively low power to switch their resistance states, which can contribute to energy-efficient operation in integrated circuits. RRAM devices have the potential to offer fast read and write access times compared to some other non-volatile memory technologies, making them suitable for applications requiring quick data retrieval. RRAM that can be integrated in advanced semiconductor manufacturing processes is beneficial to integration into modern microchips without major modifications to an existing fabrication process. A planar memory device may include two separate planar devices, including a single transistor (1T) and a single resistor (1R) that are typically positioned in two separate metal layers. Hence, device density increases become difficult.

[0017] In the embodiments, a grounded resistor or “ground fused device” is stacked on a source/drain to reduce passive device area for further device density improvement. An RRAM device in accordance with various embodiments may be a vertical RRAM device, and may be referred to as the vertical RRAM device. For example, the RRAM device may be used with front-end-of-line (FEOL) planar, fin-type FET or gate-all-around FET (GAAFET) logic devices. Interconnects between the RRAM device and the logic device(s) may be via a back-end-of-line (BEOL) metal routing. The vertical RRAM device may be inserted as a vertical gate-all-around RRAM device between two metal layers in an interconnect structure (e.g., BEOL or backside interconnect), such as between an M_x and M_{x+1} metal layer (e.g., M3 and M4, M6 and M7, or the like). The vertical RRAM device may include a metal-insulator-metal (MIM) fused device on top of or underneath an oxide semiconductor channel, such as an indium-gallium-zinc-oxide (IGZO) channel.

[0018] Nanostructure transistor structures may be patterned by any suitable method. For example,

the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the nanostructure structure.

[0019] FIGS. 1A-1R are views of various embodiments of memory devices **20**, **20A**, **20M** at various stages of fabrication according to aspects of the present disclosure. FIGS. 2A and 2B are diagrammatic cross-sectional and circuit diagram views of various embodiments of IC devices **10**, **10A** including the memory device **20**, **20A**, **20M** according to aspects of the present disclosure. FIG. 5 is a flowchart of a method **1000** of forming a memory device in accordance with various embodiments.

[0020] FIG. 5 illustrates a flowchart of method **1000** for forming an IC device or a portion thereof from a workpiece, according to one or more aspects of the present disclosure. Method **1000** is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated in method **1000**. Additional acts can be provided before, during and after the method **1000** and some acts described can be replaced, eliminated, or moved around for additional embodiments of the method. Not all acts are described herein in detail for reasons of simplicity. Method **1000** is described below in conjunction with fragmentary perspective and/or cross-sectional views of a workpiece, shown in FIGS. 1A-1R, at different stages of fabrication according to embodiments of method **1000**. For avoidance of doubt, throughout the figures, the X direction is perpendicular to the Y direction and the Z direction is perpendicular to both the X direction and the Y direction. It is noted that, because the workpiece may be fabricated into a semiconductor device, the workpiece may be referred to as the semiconductor device as the context requires.

[0021] In FIG. 1A, a first electrode layer **162** is formed on a first dielectric layer **160** of an integrated circuit or integrated chip (IC) device. The first dielectric layer **160** may be formed on or may be included in a first portion of an interconnect structure that is on or beneath a device layer, which will be described in greater detail with reference to FIGS. 2A and 2B. Briefly, the device layer may include one or more integrated devices, such as planar FETs, fin-type FETs, nanostructure FETs, metal-oxide-semiconductor (MOS) capacitors and the like. The interconnect structure may be on the device layer, and may include a mid-end-of-line (MEOL) and/or back-end-of-line (BEOL) interconnect structure that provides electrical connection between the integrated devices of the device layer and to other device external to the IC device. The interconnect structure may include instead or additionally a backside interconnect structure on a back side of the device layer. Examples of frontside and backside interconnect structures **700**, **800** and planar FET and nanostructure FET device layers **710**, **710A** are depicted in FIGS. 2A and 2B, respectively. Acts **1010** and **1020** of method **1000**, which include forming a device layer (act **1010**) and forming a first portion of an interconnect structure on the device layer (act **1020**) are described in greater detail with reference to FIGS. 2A and 2B.

[0022] The first dielectric layer **160** may be or include an oxide, such as silicon oxide, and may be included in an interconnect structure over or under a device layer. The first dielectric layer **160** may be formed by an appropriate deposition operation, such as chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD), atomic layer deposition (ALD), or the like. The first dielectric layer **160** may be present on a metallization layer that includes metal features (e.g., contacts, traces, wires, vias, or the like) embedded in an intermetal dielectric (IMD). The IMD may be or include silicon dioxide, carbon doped silicon dioxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), a porous dielectric material, or the like.

[0023] The first electrode layer **162** is formed on the first dielectric layer **160**. In some embodiments, the first electrode layer **162** is or includes a transition metal nitride. For example, the first electrode layer **162** may be or include TiN, TaN, WN, HfN or the like, and may be formed by an appropriate deposition operation, such as a physical vapor deposition (PVD), CVD, low-pressure CVD (LPCVD), PECVD, ALD or the like.

[0024] After forming the first electrode layer **162** on the first dielectric layer **160**, an oxide semiconductor layer **164L** may be formed on the first electrode layer **162**, corresponding to act **1030** of method **1000**. The oxide semiconductor layer **164L** may be or include indium-gallium-zinc-oxide (IGZO). In some embodiments, the oxide semiconductor layer **164L** is or includes one or more of IGZO, zinc-tin-oxide (ZTO), aluminum-zinc-oxide (AZO), hafnium-indium-zinc-oxide (HIZO), gallium-zinc-oxide (GZO), indium-gallium-oxide (IGO), zinc-indium-tin-oxide (ZITO), combinations thereof and the like. The oxide semiconductor layer **164L** may be formed by an appropriate deposition operation, such as a PVD, CVD, ALD, e-beam evaporation or the like.

[0025] In FIG. **1B**, following formation of the oxide semiconductor layer **164L**, a patterned oxide semiconductor layer **164L'** including a channel **164C** is formed by patterning the oxide semiconductor layer **164L**. The channel **164C** may have the shape of a pillar. Namely, the channel **164C** may have cross-sectional profile in the XY plane that is circular, square, rectangular, an irregular shape, or the like. A height of the channel **164C** in the Z-axis direction over a width of the channel **164C** in the X-axis or Y-axis direction may be referred to as an aspect ratio of the channel **164C**. The aspect ratio of the channel **164C** may be in a range of about 1.5 to about 20, about 2 to about 15, about 4 to about 10, or another suitable range. An aspect ratio that exceeds about 20 may lead to collapse of the channel **164C**. An aspect ratio that is below about 1.5 may not provide sufficient channel length.

[0026] In FIGS. **1C-1F** a gate structure **16** (see FIG. **1F**) is formed that wraps around the channel **164C**, corresponding to act **1040** of method **1000**. The gate structure **16** includes a dielectric layer **166** and a gate conductive layer **168**.

[0027] In FIG. **1C**, following formation of the channel **164C** that has the pillar shape, a gate dielectric layer **166L** is formed on the oxide semiconductor layer **164L'** including the channel **164C**. The gate dielectric layer **166L** wraps around and covers the channel **164C**. Namely, the gate dielectric layer **166L** may be in direct contact with sidewalls and an upper surface of the channel **164C**. The gate dielectric layer **166L** may be or include at least one dielectric material, such as SiO₂, or a high-k gate dielectric material, which may refer to dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide ($k \approx 3.9$). Example high-k dielectric materials include HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, ZrO₂, Ta₂O₅, or combinations thereof. In some embodiments, the gate dielectric layer **166L** includes one or more of SiO₂, HfO, La, SiON, SiCON, Zn or Zr. In some embodiments, the gate dielectric layer **166L** has thickness of about 5 Angstroms (Å) to about 100 Å. The high-k dielectric material(s) may be deposited by ALD, CVD, PVD, molecular beam epitaxy (MBE), or the like.

[0028] Further to FIG. **1C**, following formation of the gate dielectric layer **166L**, the gate conductive layer **168L** is formed on the gate dielectric layer **166L**. The gate conductive layer **168L** may be in direct contact with the gate dielectric layer **166L**. For example, the gate conductive layer **168L** may be in direct contact with sidewalls and upper surfaces of the gate dielectric layer **166L**. The gate conductive layer **168L** may include a conductive material such as polysilicon, silicon, titanium, tantalum, aluminum, tungsten, nickel, zinc, indium, gallium, germanium, carbon, cobalt, ruthenium, iridium, molybdenum, copper, or combinations thereof. The conductive material may be deposited by an appropriate deposition operation, such as a CVD, PVD, ALD, e-beam evaporation, or the like. In some embodiments, the gate conductive layer **168L** has thickness of about 5 Å to about 100 Å.

[0029] In FIG. **1D**, following formation of the gate conductive layer **168L**, a mask layer **170**, which may be or include an oxide layer, is formed on the structure depicted in FIG. **1C**. The oxide layer

170 may be a blanket layer that covers the gate conductive layer **168L**. The oxide layer **170** may be formed by any of the methods described above with reference to the oxide layer **160**.

[0030] In FIG. **1E**, following deposition of the oxide layer **170**, the oxide layer **170** is patterned to expose an electrode contact portion **162C**, highlighted in FIG. **1E** in phantom. After patterning the oxide layer **170**, exposed portions of the gate conductive layer **168L**, the gate dielectric layer **166L** and the oxide semiconductor layer **164L'** are removed to expose the electrode contact portion **162C**. The oxide layer **170** may be etched by reactive ion etching (RIE), wet etching, plasma etching, inductively coupled plasma (ICP) etching, or the like. Following patterning of the oxide layer **170**, exposed portions of the gate conductive layer **168L** may be etched, for example, by an RIE, ICP etch, or the like, that uses an etchant such as CF₄, resulting in a gate conductive layer **168L'**. Then, the gate dielectric layer **166L** may be etched, for example, by a dry etch, such as RIE, ICP etch, or the like, that uses a chlorine- or fluorine-based etchant, resulting in a gate dielectric layer **166L'**. Then, the oxide semiconductor layer **164L'** may be etched, for example, by a dry etch, such as an RIE, ICP etch or the like, that uses a chlorine- or fluorine-based etchant, resulting in a channel layer **164**.

[0031] In FIG. **1F**, following formation of the gate conductive layer **168L'**, the gate dielectric layer **166L'** and the channel layer **164**, portions of the gate conductive layer **168L'** and the gate dielectric layer **166L'** on an upper region of the channel **164C** are removed. In some embodiments, following the operations of FIG. **1F**, the oxide layer **170** is removed and another oxide layer **172** is formed. The oxide layer **172** is then recessed to a level that is below an upper surface of the channel **164C**, for example, by a chemical mechanical planarization (CMP), anisotropic etch, combination thereof, or the like. After recessing, the portions of the gate conductive layer **168L'** and the gate dielectric layer **166L'** on the upper region of the channel **164C** are exposed. In some embodiments, a portion of the channel **164C** that protrudes from the oxide layer **172** has height that is in a range of about 1/10^{sup.th} to about 2/3^{sup.rd} total height of the channel **164C**.

[0032] Following recessing of the oxide layer **172**, exposed portions of the gate conductive layer **168L'** and the gate dielectric layer **166L'** on the upper region of the channel **164C** are removed. The removal may be similar to that described with reference to FIG. **1E**, including similar etching processes and chemistries. Following removal of the exposed portions, a gate conductive layer **168** and a gate dielectric layer **166** remain, the resulting structure being depicted in FIG. **1F**. The gate dielectric layer **166** includes vertical portion(s) **166V** that wrap around the channel **164C** and horizontal portions **166H** that cover horizontal portions **164H** of the oxide semiconductor layer **164**. The gate conductive layer **168** includes vertical portion(s) **168V** that wrap around the channel **164C** and horizontal portions **168H** that cover horizontal portions **166H**, **164H** of the gate dielectric layer **166** and the oxide semiconductor layer **164**, respectively.

[0033] In FIGS. **1G-11**, a fused device **18** is formed on the exposed upper portion of the channel **164C**, corresponding to act **1050** of method **1000**. The fused device **18** may be a MIM device. The fused device **18** may be a memristor, and may include one or more of the following layers: bottom electrode, active or switching layer (insulator), barrier or buffer or ion-conductive layer (optional), top electrode, passivation layer (optional), encapsulation (optional).

[0034] In FIG. **1G**, an insulating layer **174** is formed that covers the gate dielectric layer **166** and the gate conductive layer **168**. The insulating layer **174** provides electrical isolation between the gate structure **16** and the fused device **18** that is formed in subsequent operations. In some embodiments, the insulating layer **174** is a dielectric layer that includes one or more of SiN, SiCN, SiON, SiOCN, SiOC, or the like. The insulating layer **174** may be formed by blanket depositing a conformal layer of one or more of the mentioned dielectrics on the structure depicted in FIG. **1F**. The deposition may be by a suitable deposition operation, such as a CVD, LPCVD, PECVD, ALD, PVD, MBE or the like. The insulating layer **174** may be recessed following deposition so as to expose an upper region of the channel **164C**.

[0035] In FIG. **1H**, following deposition of the insulating layer **174**, material layers for forming the

fused device **18** may be deposited in sequence as blanket conformal layers that cover the channel **164C** and the insulating layer **174**.

[0036] First, a bottom electrode layer **180L** may be formed. The bottom electrode layer **180L** may include a transition metal nitride, such as TiN, or another conductive material. The bottom electrode layer **180L** may be formed by a suitable deposition operation, such as an ALD, CVD, or the like.

[0037] Following formation of the bottom electrode layer **180L**, an active or switching layer **182L** may be formed. The switching layer **182L** may be a high-k dielectric layer, and may include at least one of dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide ($k \approx 3.9$). Example high-k dielectric materials include TiO₂, HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, ZrO₂, Ta₂O₅, or combinations (e.g., multilayers) thereof. The switching layer **182L** may be formed by a suitable deposition operation, such as an ALD, CVD or the like.

[0038] Following formation of the switching layer **182L**, an optional barrier, buffer or ion-conducting layer **184L** may be formed. The ion-conducting layer **184L** may also be referred to as an “oxygen reservoir layer.” The optional layer **184L** may be a layer of titanium or tantalum. The titanium or tantalum layer may be formed by a suitable deposition operation on the switching layer **182L**, such as by an ALD, CVD or the like.

[0039] Following formation of the optional layer **184L** or the switching layer **182L**, a top electrode layer **186L** may be formed thereon. The top electrode layer **186L** may include a transition metal nitride, such as TiN, or another conductive material. The top electrode layer **186L** may be formed by a suitable deposition operation, such as an ALD, CVD, or the like. In some embodiments, the top electrode layer **186L** is the same material or substantially the same material as the bottom electrode layer **180L**.

[0040] In FIG. 1I, following formation of the bottom electrode layer **180L**, the switching layer **182L**, the optional barrier, buffer or ion-conducting layer **184L** and the top electrode layer **186L**, a patterning operation is performed to form the fused device **18**. The patterning may include one or etching operations that remove some portions of each of the bottom electrode layer **180L**, the switching layer **182L**, the optional barrier, buffer or ion-conducting layer **184L** and the top electrode layer **186L** between neighboring channels **164C** so as to form individual fused devices **18** over each channel **164C** that are electrically and physically isolated from each other. In addition, removing material of the bottom electrode layer **180L**, the switching layer **182L**, the optional barrier, buffer or ion-conducting layer **184L** and the top electrode layer **186L** between the channels **164C** opens up space for forming contacts to the first electrode layer **162** and the gate conductive layer **168**, which is described in greater detail with reference to FIGS. 1L-1N below. Removal of the bottom electrode layer **180L**, the switching layer **182L**, the optional barrier, buffer or ion-conducting layer **184L** and the top electrode layer **186L** may be by RIE, ICP etch, or the like. A resulting structure in which the fused device **18** includes a bottom electrode **180**, a switching layer **182**, an optional barrier, buffer or ion-conducting layer **184** and a top electrode **186** is depicted in FIG. 1I.

[0041] In FIGS. 1J and 1K, spacers **188U**, **188L** are formed on the fused device **18**, corresponding to act **1060** of method **1000**. The spacers **188U**, **188L** include an upper spacer **188U** and a lower spacer **188L** that wrap around upper and lower portions of the fused device **18**, respectively. The spacers **188U**, **188L** are beneficial to provide protection during formation of contacts, and to provide physical isolation and electrical isolation between the fused device **18** and the contacts after formation.

[0042] In FIG. 1J, a spacer material layer **188M** is formed on the fused devices **18** and the insulating layer **174**. The spacer material layer **188M** may be or comprise one or more of SiN, SiCN, SiC, SiOC, SiOCN, HfO₂, ZrO₂, ZrAlO_x, HfAlO_x, HfSiO_x, Al₂O₃, or other suitable material. The spacer material layer **188M** may be formed by a suitable deposition operation, such as a LPCVD, PECVD, high-density plasma CVD (HDPCVD), ALD, or the like. The spacer

material layer **188M** may be blanket deposited as a conformal thin layer on the fused device **18** and exposed regions of the insulating layer **174**, resulting in the structure shown in FIG. **1J**.

[0043] In FIG. **1K**, the spacer material layer **188M** is etched to remove horizontal portions thereof overlying the top electrode **186** and the insulating layer **174**. The etching may include RIE, ICP, wet etching or another suitable etch operation. Following etching, upper and lower spacers **188U**, **188L** remain on the fused device **18**, as depicted in FIG. **1K**. The upper spacer **188U** may wrap around an upper region of the fused device **18**, and may be positioned on horizontal portions of the top electrode **186**. The lower spacer **188L** may wrap around a lower region of the fused device **18**, may be positioned adjacent (and in direct contact with) sidewalls or terminuses of horizontal portions of the top electrode **186**, the ion-conducting layer **184**, the switching layer **182** and the bottom electrode **180**, and may be positioned on an upper surface of the insulating layer **174**. The upper and lower spacers **188U**, **188L** may each have a rounded profile. Namely, outer sidewalls thereof may be rounded instead of vertical following the etching operation.

[0044] In FIGS. **1L-1N**, contacts **194**, **196**, **198** are formed that land on the electrode contact portion **162C**, the top electrode **186** and the gate conductive layer **168**, respectively, corresponding to act **1070** of method **1000**. The contacts **194**, **196**, **198** provide electrical connection to the RRAM device **20**. For example, a bit line (BL), word line (WL) and ground (or other bias voltage) may be connected to the RRAM device **20** via the contacts **194**, **196**, **198**, which may carry voltages that are applied to the electrode contact portion **162C**, the top electrode **186** and the gate conductive layer **168** to program, erase or otherwise operate the RRAM device **20**. FIG. **1O** is a simple circuit schematic diagram of the RRAM device **20** in accordance with various embodiments.

[0045] In FIG. **1L**, a mask layer **500** is formed over the fused device **18** and the insulating layer **174**. The mask layer **500** may be or comprise an oxide layer, such as a silicon oxide, and may be deposited by any of the methods described above for the mask layer **170**.

[0046] In FIG. **1M**, openings **190BL**, **190G**, **190WL** are formed in the mask layer **500** and the underlying structure to expose the electrode contact portion **162C**, the top electrode **186** and the gate conductive layer **168**. For example, the openings **190BL**, **190G**, **190WL** may be formed initially through the mask layer **500** and land on the insulating layer **174** and the top electrode **186**. Then, the openings **190BL**, **190WL** may be extended through the insulating layer **174** and the oxide layer **172** to expose the electrode contact portion **162C** and the gate conductive layer **168**. The openings **190BL**, **190G**, **190WL** may be formed by one or more appropriate etch operations, such as an RIE, ICP, or the like. The remaining mask layer **500** is referred to as oxide layer **192**.

[0047] In FIG. **1N**, a source/drain contact **194**, a gate contact **198** and a top electrode contact **196** (or collectively “the contacts **194**, **196**, **198**”) are formed in the openings **190BL**, **190WL**, **190G**, respectively. The contacts **194**, **196**, **198** may be or include one or more of a liner layer, barrier layer, glue layer, conductive core layer, and the like. In some embodiments, the contacts **194**, **196**, **198** include one or more of Cu, Co, Al, Ni, W, Ru, Ti, TiN, Ta, TaN, alloys thereof, multilayers thereof, combinations thereof, or the like. The layer(s) of the contacts **194**, **196**, **198** may be formed by one or more appropriate deposition operations, such as a PVD, CVD, ALD, or the like.

[0048] In FIG. **1O**, the RRAM device **20** may be represented in simple circuit form as a 1T1R (“one transistor, one resistor”) circuit that includes a transistor **260** and a resistor **270**, which may be a memristor **270**. The gate conductive layer **168**, the gate dielectric layer **166** and the channel **164C** may be included in a transistor **260**. The gate conductive layer **168** or gate electrode of the transistor **260** may be connected to a word line WL via the contact **198**. A source/drain electrode of the transistor **260**, which may be the electrode layer **162**, may be connected to a bit line BL via the contact **194**. The top electrode **186** of the fused device **18**, which may be the memristor **270**, may be connected to a bias voltage (e.g., ground) via the contact **196**.

[0049] Following formation of the contacts **194**, **196**, **198**, a second portion of the interconnect structure may be formed on the contacts **194**, **196**, **198**, corresponding to act **1080** of method **1000**. A front side interconnect structure **700** and a back side interconnect structure **800** including a first

portion and a second portion with an RRAM device **20** therebetween are depicted in FIG. 2A and FIG. 2B.

[0050] FIGS. 1P-1R are diagrams that illustrate fused devices **18A**, **18M** in accordance with various embodiments. In FIG. 1P, the fused device **18A** has an angular shape. In FIGS. 1Q and 1R, the fused device **18M** has a cross-sectional profile that is M-shaped.

[0051] In FIG. 1P, the fused device **18A** has an angular or sharp-tipped cross-sectional profile in the XZ plane and/or the YZ plane. The angular shape may be due to etching of the upper portion of the channel **164C** that protrudes above the insulating layer **174**. In some embodiments, after etching the upper portion, the upper portion of the channel **164C** has a conical or pyramidal shape in perspective view, such that the subsequent layers formed thereon inherit the conical or pyramidal shape, as depicted in FIG. 1P. Namely, each of the bottom electrode **180A**, the switching layer **182A**, the ion-conducting layer **184A** and the top electrode **186A** may have tapered sidewalls that meet in respective points above the channel **164C**. Due to the angular or sharp-tipped profile of the top electrode **186A**, contact area between the contact **196** and the top electrode **186A** may be increased, which may be beneficial to reduce contact resistance therebetween and improve performance of the RRAM device **20A** that includes the fused device **18A**.

[0052] In FIG. 1Q, the fused device **18M** has an M-shaped cross-sectional profile in the XZ plane and/or the YZ plane. The M shape may be due to etching of the upper portion of the channel **164C** that protrudes above the insulating layer **174**. Namely, a center region of the upper portion may be recessed to have an inverted cone or inverted pyramid cutout shape in perspective view. In some embodiments, after etching the upper portion, the subsequent layers formed thereon inherit the inverted cone or inverted pyramid shape, as depicted in FIG. 1Q. Namely, each of the bottom electrode **180M**, the switching layer **182M**, the ion-conducting layer **184M** and the top electrode **186M** may have tapered sidewalls that meet in respective points within the channel **164C**. Due to the angular or sharp-tipped profile of the top electrode **186M**, contact area between the contact **196** and the top electrode **186M** may be increased, which may be beneficial to reduce contact resistance therebetween and improve performance of the RRAM device **20M** that includes the fused device **18M**.

[0053] FIG. 1R is a detailed view of the fused device **18M** of the RRAM device **20M** of FIG. 1Q in accordance with various embodiments. As depicted, each of the bottom electrode **180M**, the switching layer **182M**, the ion-conducting layer **184M** and the top electrode **186M** may include horizontal portions **180B**, **182B**, **184B**, **186B**, vertical sidewall portions **180S**, **182S**, **184S**, **186S** and V-shaped portions **180V**, **182V**, **184V**, **186V**, respectively. The V-shaped portions **180V**, **182V**, **184V**, **186V** are V-shaped in cross-section, but may be conical or pyramidal in perspective view.

[0054] The V-shaped portion **180V** may increase contact area between the channel **164C** and the bottom electrode **180M**, which may improve (e.g., reduce) contact resistance therebetween and improve device performance. Namely, in addition to being in contact (e.g., direct contact) with the channel **164C** via the vertical sidewall portion **180S**, contact area between the upper surface of the channel **164C** and the bottom electrode **180M** is increased by the V-shaped portion **180V** that extends into the channel **164C**. In some embodiments, the V-shaped portion **180V** of the bottom electrode **180M** may extend below the horizontal portion **180B** thereof by a distance **D1**. The distance **D1** may be in a range of about 10 Å to about 10 nm. In some embodiments, the V-shaped portion **180V** extends to the same level as the lower surface of the horizontal portion **180B** or to a level that is above the lower surface or the upper surface of the horizontal portion **180B**. Although the V-shaped portion **180V** is depicted as having a sharp tip, in some embodiments, the V-shaped portion **180V** may instead have a rounded or blunted tip. In some embodiments, the V-shaped portion **180V** is not formed and instead a U-shaped portion is formed.

[0055] FIGS. 2A and 2B are diagrammatic cross-sectional views of the RRAM device **20** included in IC devices **10**, **10A** in accordance with various embodiments. In FIG. 2A, the RRAM device **20** is included in a front side interconnect structure **700**. In FIG. 2B, the RRAM device **20** is included

in a back side interconnect structure **800**. It should be understood that some embodiments of an IC device may include the RRAM device **20** in both a front side interconnect structure and a back side interconnect structure, which may be beneficial to increase density of RRAM devices per unit area of the IC device. It should also be understood that planar transistors **300** are depicted in FIG. 2A and nanostructure transistors **80** (e.g., GAAFETs) are depicted in FIG. 2B only for purposes of illustration. The nanostructure transistors **80** can also be used in the arrangement depicted in FIG. 2A and the planar transistors **300** can also be used in the arrangement depicted in FIG. 2B. Other transistor types (e.g., FinFETs) may be included instead of the planar transistors **300** and the nanostructure transistors **80** in the IC devices **10**, **10A** in accordance with various embodiments. [0056] In FIG. 2A, the IC device **10** includes a substrate **110**, a device layer **710** on and/or in the substrate **110** and a front side interconnect structure **700** on the device layer **710** that includes the RRAM device **20**. The device layer **710** in FIG. 2A is depicted as including planar transistors **300**. In some embodiments, the device layer **710** may include fin-type transistors, nanostructure transistors (e.g., GAAFETs) and the like.

[0057] The substrate **110** may be a semiconductor substrate, such as a bulk semiconductor, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The semiconductor material of the substrate **110** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; other compound semiconductors including gallium, zinc, indium and/or oxygen; or combinations thereof. Other substrates, such as single-layer, multi-layered, or gradient substrates may be used.

[0058] The planar transistors **300** include source/drains **82P** and a gate structure **200P**, and may be formed on and in wells **850** in the substrate **110**.

[0059] The source/drains **82P** may be heavily doped regions of the substrate **110** in the wells **850** that are doped with impurities to form either n-type (with donor impurities like phosphorus or arsenic) semiconductor for NMOS or p-type (with acceptor impurities like boron) semiconductor for PMOS. Between the source/drain regions **82P** is a channel that is lightly doped with the opposite type of impurity compared to the source/drain regions **82P**. For example, an NMOS may have a p-type channel (e.g., with boron), and a PMOS may have an n-type channel (e.g., with phosphorus). In some embodiments, the source/drains **82P** are doped with carbon.

[0060] The gate structure **200P** may be or include polysilicon (“polycrystalline silicon”) or a metal such as tungsten. The gate structure **200P** may include an insulating layer or “gate dielectric” that may be or include silicon dioxide (SiO₂) or a high-k dielectric, such as hafnium oxide (HfO₂) that is beneficial to reduce leakage and improve performance.

[0061] Although not separately depicted in FIG. 2A, contacts that provide electrical connection to the source/drains **82P** and the gate structure **200P** may be present. The contacts may include tungsten (W), aluminum (Al) or another suitable conductive material. The contacts may include one or more barrier layers, such as titanium nitride, and adhesion layers that may be used to improve physical contact and reduce diffusion.

[0062] The front side interconnect structure **700** is on the gate structure **200P**, the source/drains **82P** and the substrate **110**. The front side interconnect structure **700** may include conductive features **730** embedded in dielectric layers **720**. For example, the front side interconnect structure **700** may include one or more metal layers (e.g., M₀, M₁, M₂, . . . , M_{sub.N}), each of which includes the dielectric layer **720**, which may be an intermetal dielectric (IMD) layer. Each IMD layer may be or include one or more of silicon oxynitride, phosphosilicate glass (PSG), borosilicate glass (BSG), borophosphosilicate glass (BPSG), undoped silicate glass (USG), fluorinated silicate glass (FSG), silicon oxycarbide (SiO_xC_y), Spin-On-Glass (SOG) or a combination thereof. Conductive features **730** may be connected to each other by conductive vias **740**. For example, a first conductive feature

730A may be connected to a second conductive feature 730B by a conductive via 740. The conductive features 730, 730A, 730B and the conductive vias 740 may be or include one or more of a metal, such as copper, aluminum, TiN, TaN, Ta, graphene, carbon nanotubes, conductive polymer, combinations thereof and the like. The conductive feature 730 is electrically connected to the gate structure 200 via a conductive feature 210.

[0063] The RRAM device 20 may be positioned between two or more of the metal layers. For example, the RRAM device 20 may be positioned between a third metal layer (M3) and a fourth metal layer (M4). In some embodiments, the RRAM device 20 is positioned between two metal layers that are more than one level apart. For example, the RRAM device 20 may be positioned between a fourth metal layer (M4) and a sixth metal layer (M6) or seventh metal layer (M7).

[0064] In FIG. 2B, the RRAM device 20 is positioned in a back side interconnect structure 800. The back side interconnect structure 800 is similar in many respects to the front side interconnect structure 700 of FIG. 2A, but may be formed on a backside of a device layer 710A. For example, the substrate 110 may be removed partially or entirely, and contact to source/drains 82 may be from a back side of the source/drains 82, as depicted in FIG. 2A. The back side interconnect structure 800 may include conductive features 820 embedded in dielectric layer 810. A front side interconnect structure 700A is depicted in FIG. 2B. The front side interconnect structure 700A is similar in most respects to the front side interconnect structure 700 and includes conductive features 730 embedded in dielectric layers 720. In some embodiments, a semiconductor layer 900 is positioned on the front side interconnect structure 700A opposite the device layer 710A. The semiconductor layer 900 may be undoped silicon.

[0065] FIG. 2B also depicts the device layer 710A including nanostructure devices 80 instead of planar devices 300. A single nanostructure device 80 is described in the following. The nanostructure devices 80 may be or include one or more N-type FETs (NFETs) or P-type FETs (PFETs). Nanostructure devices 80 may be separated (e.g., physically and/or electrically isolated) from each other by shallow trench isolation (STI), deep trench isolation (DTI), local oxidation of silicon (LOCOS), or the like.

[0066] The nanostructure device 80 is formed over and/or in a substrate 110, and generally includes a gate structure 200 that straddles and/or wraps around semiconductor channels or nanostructures 22. The gate structure 200 controls electrical current flow through the channels 22.

[0067] The nanostructure device 80 is shown including four channels 22, which are laterally abutted by source/drain features or regions 82 and covered and surrounded by the gate structure 200. Generally, the number of channels 22 is two or more, such as three or four or more. The gate structure 200 controls flow of electrical current through the channels 22 to and from the source/drain features 82 based on voltages applied at the gate structure 200 and at the source/drain features 82.

[0068] In some embodiments, the nanostructure device includes an NFET, and the source/drain features 82 thereof include silicon phosphorous (SiP), SiAs, SiSb, SiPAs, SiP:As:Sb, combinations thereof, or the like. In some embodiments, the nanostructure device 20 includes a PFET, and the source/drain features 82 thereof include silicon germanium (SiGe), either undoped or doped to form, for example, SiGe:B, SiGe:B:Ga, SiGe:Sn, SiGe:B:Sn, or another appropriate semiconductor material. Generally, the source/drain features 82 may include any combination of appropriate semiconductor material(s) and appropriate dopant(s).

[0069] The channels 22 each include a semiconductive material, for example silicon or a silicon compound, such as silicon germanium, or the like. The channels 22 are nanostructures (e.g., having sizes that are in a range of a few nanometers) and may also each have an elongated shape and extend in the X-direction. In some embodiments, the channels 22 each have a nanowire (NW) shape, a nanosheet (NS) shape, a nanotube (NT) shape, or other suitable nanoscale shape. The cross-sectional profile of the channels 22 may be rectangular, round, square, circular, elliptical, hexagonal, or combinations thereof.

[0070] In some embodiments, the lengths (e.g., measured in the X-direction) of the channels **22** may be different from each other, for example due to tapering during a fin etching process. The channels **22** each may not have uniform thickness (e.g., along the X-axis direction), for example due to a channel trimming process used to expand spacing (e.g., measured in the Z-axis direction) between the channels **22** to increase gate structure fabrication process window. For example, a middle portion of each of the channels **22** may be thinner than the two ends of each of the channels **22**. Such shape may be collectively referred to as a “dog-bone” shape.

[0071] In some embodiments, the spacing between the channels **22** in the vertical direction is in a range between about 8 nanometers (nm) and about 12 nm. In some embodiments, a thickness (e.g., measured in the Z-direction) of each of the channels **22** is in a range between about 5 nm and about 8 nm. In some embodiments, a width (e.g., measured in the Y-direction) of each of the channels **22** is at least about 8 nm.

[0072] The gate structure **200** is disposed over and between the channels **22** respectively. In some embodiments, the gate structure **200** is disposed over and between the channels **22** which are silicon channels for N-type devices or silicon germanium channels for P-type devices. In some embodiments, the gate structure **200** includes an interfacial layer (IL), one or more gate dielectric layers, one or more work function tuning layers and a metal core layer.

[0073] The interfacial layer, which may be an oxide of the material of the channels **22** is formed on exposed areas of the channels **22**. The interfacial layer promotes adhesion of the gate dielectric layers to the channels **22**. In some embodiments, the interfacial layer has thickness of about 5 Angstroms (Å) to about 50 Angstroms (Å). In some embodiments, the interfacial layer has thickness of about 10 Å. The interfacial layer having thickness that is too thin may exhibit voids or insufficient adhesion properties. The interfacial layer being too thick consumes gate fill window, which is related to threshold voltage tuning and resistance. In some embodiments, the interfacial layer is doped with a dipole, such as lanthanum, for threshold voltage tuning.

[0074] In some embodiments, the gate dielectric layer includes at least one high-k gate dielectric material, which may refer to dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide ($k \approx 3.9$). Example high-k dielectric materials include HfO_2 , HfSiO , HfSiON , HfTaO , HfTiO , HfZrO , ZrO_2 , Ta_2O_5 , or combinations thereof. In some embodiments, the gate dielectric layer has thickness of about 5 Å to about 100 Å. The gate dielectric layer may be a single layer or a multilayer that is formed on the IL.

[0075] The gate structure **200** also includes metal core layer on the gate dielectric layer. The metal core layer may include a conductive material such as Co, W, Ru, combinations thereof, or the like. In some embodiments, the metal core layer is or includes a Co-, W- or Ru-based compound or alloy including one or more elements, such as Zr, Sn, Ag, Cu, Au, Al, Ca, Be, Mg, Rh, Na, Ir, W, Mo, Zn, Ni, K, Co, Cd, Ru, In, Os, Si, Ge, Mn, combinations thereof, or the like. Between the channels **22**, the metal core layer is circumferentially surrounded (in the cross-sectional view) by the one or more work function metal layers, which are then circumferentially surrounded by the gate dielectric layers, which are circumferentially surrounded by the interfacial layer. The gate structure **200** may also include a glue layer that is formed between the one or more work function layers and the metal core layer to increase adhesion. The glue layer is not specifically illustrated in FIG. 2B for simplicity.

[0076] The nanostructure device **80** may include gate spacers **41** that are disposed on sidewalls of the metal core layer, the gate dielectric layer and the IL above the uppermost channel **22**, and inner spacers **74** that are disposed on sidewalls of the IL and/or the gate dielectric layer between the channels **22**. The inner spacers **74** are also disposed between the channels **22**. In the embodiment depicted in FIG. 2B, the gate spacers **41** include two spacer layers. In some embodiments, the gate spacers **41** include a single spacer layer or three or more spacer layers. The first and second spacer layers may each include a dielectric material, for example a low-k material such as SiOCN , SiON , SiN , SiCN , SiOC or the like. In some embodiments, the second spacer layer is not present.

Material of the first and second spacer layers may be the same as or different from each other. Generally, an upper portion of the second spacer layer (or the first spacer layer when the second spacer layer is not present) may be removed partially or fully to increase aspect ratio of an opening through which the source/drain region **82** is formed.

[0077] Contact to the source/drains **82** from the back side of the device layer **710A** may be by a conductive back side via **840**. The back side via **840** may be in contact with one or more of the source/drains **82** and with a conductive feature **820** in a back side metal layer (e.g., “BM0,” “BM1,” or the like).

[0078] The RRAM device **20** may be positioned between two of the back side metal layers, such as between a second back side metal layer (BM2) and a third or fourth back side metal layer (BM3, BM4). A conductive via **830** may extend adjacent the RRAM device **20** and may connect a conductive feature **820A** of a metal layer above the RRAM device **20** to a conductive feature **820B** of a metal layer below the RRAM device **20**. “Above” and “below” in this description may refer to orientation of the page and are interchangeable based on orientation of the IC device **10A**. Namely, when the back side of the IC device **10A** is facing up, as depicted in FIG. 2B, the conductive feature **820A** is “above” the conductive feature **820B**.

[0079] In some embodiments, the conductive feature **820B** is a power rail of the back side interconnect structure **800**. The back side interconnect structure **800** may include at least two power rails **820B**, **820C**. A first power rail **820C** connects to logic devices (e.g., the transistor **80**) and a second power rail **820B** connects to the RRAM device **20**. The first power rail **820C** may have dimension that is equal to or smaller than dimension of the second power rail **820B**. For example, the second power rail **820B** that is used for driving the RRAM device **20** may carry a voltage that exceeds voltage carried by the first power rail **820C**. The larger dimension(s) of the second power rail **820B** may be beneficial to improve reliability under the higher voltage carried thereon. In some embodiments, the power rails **820B**, **820C** are or include W, Ru, Ir, Mo or the like.

[0080] In the embodiments described with reference to FIGS. 1A-2B, the contact **194** that connects the bit line BL to the drain of the transistor **260** via the electrode layer **162** is connected to an upper metal layer. In some embodiments, instead of connecting to the upper metal layer as depicted in FIGS. 2A and 2B, the contact **194** may be formed prior to formation of the RRAM device **20** such that the electrode layer **162** may be connected to a lower metal layer. In such an embodiment, the operation depicted in FIG. 1E may be omitted.

[0081] FIGS. 3A-3T are views of various embodiments of an RRAM device **40** of at various stages of fabrication according to aspects of the present disclosure. FIG. 3U is a circuit schematic diagram of the RRAM device **40**. FIGS. 4A and 4B are diagrammatic cross-sectional views of various embodiments of IC devices **10**, **10A** according to aspects of the present disclosure. FIG. 6 is a flowchart of a method **2000** of forming an IC device in accordance with various embodiments.

[0082] FIG. 6 illustrates a flowchart of method **2000** for forming an IC device or a portion thereof from a workpiece, according to one or more aspects of the present disclosure. Method **2000** is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated in method **2000**. Additional acts can be provided before, during and after the method **2000** and some acts described can be replaced, eliminated, or moved around for additional embodiments of the method. Not all acts are described herein in detail for reasons of simplicity. Method **2000** is described below in conjunction with fragmentary perspective and/or cross-sectional views of a workpiece, shown in FIGS. 3A-3T, at different stages of fabrication according to embodiments of method **2000**. For avoidance of doubt, throughout the figures, the X direction is perpendicular to the Y direction and the Z direction is perpendicular to both the X direction and the Y direction. It is noted that, because the workpiece may be fabricated into a semiconductor device, the workpiece may be referred to as the semiconductor device as the context requires.

[0083] In FIGS. 3A-3T, a fused device **38** is formed prior to forming a transistor of the RRAM device **40**. Some operations of the method **2000** are similar in many respects to those of method

1000 of FIG. 5, and reference to the related description will be provided.

[0084] In FIG. 3A, a first dielectric layer **360** is formed, which may be similar in most respects to the first dielectric layer **160** of FIG. 1A. Following formation of the first dielectric layer **360**, an opening **37** may be formed in the first dielectric layer **360**, in which the fused device **38** will be formed. Prior to forming the first dielectric layer **360**, a device layer of the IC device **10** may be formed, as described with reference to FIGS. 1A, 2A and 2B above, corresponding to act **2010** of method **2000**. Then, a first portion of an interconnect structure may be formed on the device layer, corresponding to act **2020** of method **2000**. Namely, the RRAM device **40** may be formed on a metal layer of an interconnect structure.

[0085] In FIGS. 3B-3E, following formation of the opening **37** in the first dielectric layer **360**, the fused device **38** is formed on the first dielectric layer **360** including in the opening **37**, corresponding to act **2030** of method **2000**. The fused device **38** may be a memristor, and may include one or more of the following layers: bottom electrode, active or switching layer (insulator), barrier or buffer or ion-conductive layer (optional), top electrode, passivation layer (optional), encapsulation (optional).

[0086] In FIG. 3B, material layers for forming the fused device **38** may be deposited in sequence as blanket conformal layers.

[0087] First, a top electrode **386** may be formed that includes a transition metal nitride, such as TiN, or another conductive material. The top electrode **386** may be formed by a suitable deposition operation, such as an ALD, CVD, or the like.

[0088] Following formation of the top electrode **386**, an optional barrier, buffer or ion-conducting layer **384L** may be formed. The optional layer **384L** may be a layer of titanium. The titanium layer may be formed by a suitable deposition operation on the top electrode **386**, such as by an ALD, CVD or the like.

[0089] Following formation of the optional ion-conducting layer **384L** or the top electrode **386**, an active or switching layer **382L** may be formed. The switching layer **382L** may be a high-k dielectric layer, and may include at least one of dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide ($k \approx 3.9$). Example high-k dielectric materials include TiO₂, HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, ZrO₂, Ta₂O₅, or combinations (e.g., multilayers) thereof. The switching layer **382L** may be formed by a suitable deposition operation, such as an ALD, CVD or the like.

[0090] Following formation of the switching layer **382L**, a bottom electrode layer **380L** may be formed. The bottom electrode layer **380L** may include a transition metal nitride, such as TiN, or another conductive material. The bottom electrode layer **380L** may be formed by a suitable deposition operation, such as an ALD, CVD, or the like. In some embodiments, the bottom electrode layer **380L** is the same material or substantially the same material as the top electrode **386**.

[0091] In FIG. 3C, a mask layer **370** is formed on the layers **380L**, **382L**, **384L**, **386**, as shown. The mask layer **370** may be similar in most respects to the mask layer **170** of FIG. 1D.

[0092] In FIG. 3D, portions of the mask layer **370** above the opening **37** are removed by a suitable removal operation, such as a CMP. Following the CMP, the upper surface of the bottom electrode layer **380L** outside the opening **37** is exposed.

[0093] In FIG. 3E, an electrode contact portion **386C** of the top electrode **386** may be exposed by exposing a portion of the top electrode **386** via a suitable etch operation(s). For example, a patterned mask may be formed that covers the top electrode **386** and the mask layer **370** and has an opening corresponding to the electrode contact portion **386C**. Exposed portions of the bottom electrode layer **380L**, the switching layer **382L** and the optional ion-conducting layer **384L** may be etched through the patterned mask as described with reference to FIG. 1I. Exposing the electrode contact portion **386C** results in a bottom electrode **380**, a switching layer **382** and an optional ion-conducting layer **384** of the fused device **38** being formed as shown.

[0094] In FIG. 3F, following exposing the electrode contact portion **386C**, a second dielectric layer **372L** is formed on the electrode contact portion **386C**, the bottom electrode **380** and the mask layer **370** in the opening **37**. The second dielectric layer **372L** may be similar in most respects to the first dielectric layer **160** described with reference to FIG. 1A.

[0095] In FIG. 3G, following formation of the second dielectric layer **372L**, a suitable removal operation, such as a CMP, is performed to remove portions of the second dielectric layer **372L** from above the bottom electrode **380**, leaving a second dielectric layer **372B** over the electrode contact portion **386C**. Upper surfaces of the second dielectric layer **372B**, the bottom electrode **380** and the mask layer **370** may be coplanar following the CMP.

[0096] In FIG. 3H, following planarization of the second dielectric layer **372B**, the bottom electrode **380** and the mask layer **370**, an insulating layer **374L** may be formed on the planarized upper surface of the second dielectric layer **372B**, the bottom electrode **380** and the mask layer **370**. The insulating layer **374L** is similar in most respects to the insulating layer **174** described with reference to FIG. 1G. The insulating layer **374L** provides electrical and physical isolation between the fused device **38** and a gate electrode layer **368** formed in a later operation.

[0097] In FIGS. 3I-3K, an oxide semiconductor channel **364** is formed on the fused device **38**, corresponding to act **2040** of method **2000**.

[0098] In FIG. 3I, an opening **35** is formed that extends through the insulating layer **374**. The opening **35** is extended by removing the mask layer **370**. In FIG. 3I, sidewalls of the insulating layer **374** and the bottom electrode **380** exposed by the opening **35** are substantially coplanar in the vertical direction. In some embodiments, the sidewalls of the insulating layer **374** are set back somewhat from the sidewalls of the bottom electrode **380**, such that at least a portion of an upper surface of the bottom electrode **380** is exposed by the opening **35**. Formation of the opening **35** may include a first etch that breaks through the insulating layer **374**, followed by a second etch that removes the mask layer **370**. The first etch may be an anisotropic etch, such as an RIE or ICP through a patterned mask. The second etch may be anisotropic or isotropic, such as an RIE, ICP, wet etch or other suitable etch that is selective to the material of the mask layer **370** without substantially attacking material of the insulating layer **374**. Profile of the opening **35** in the XY plane may be any suitable shape, such as circular, square, rectangular, or the like.

[0099] In FIG. 3J, following formation of the opening **35**, an oxide semiconductor layer **364L** is formed in the opening **35** and on the insulating layer **374**. The oxide semiconductor layer **364L** is similar in most respects to the oxide semiconductor layer **164L** described with reference to FIG. 1A, and may be or include IGZO or another suitable oxide semiconductor.

[0100] In FIG. 3K, following formation of the oxide semiconductor layer **364L**, the oxide semiconductor layer **364L** is patterned, resulting in a channel **364**, as shown. The channel **364** may include a lower portion that is below the upper surface of the insulating layer **374** and in contact with the fused device **38** and an upper portion that protrudes above the insulating layer **374**. The upper portion may have width in the X-axis direction that exceeds width of the lower portion in the X-axis direction. For example, the upper portion may partially cover an upper surface of the insulating layer **374**. The lower portion may inherit shape of the opening **35**. The upper portion may have shape (e.g., cross-sectional profile in the XY plane) that is the same as or different than that of the lower portion. For example, the lower portion may have rectangular shape and the upper portion may have circular shape.

[0101] In FIGS. 3L-3P, a gate structure **36** is formed adjacent the channel **364**, corresponding to act **2050** of method **2000**.

[0102] In FIG. 3L, a gate dielectric layer **366L** and a gate conductive layer **368L** are formed on the upper portion of the channel **364**. Formation of the gate dielectric layer **366L** and the gate conductive layer **368L** is similar in most respects to formation of the gate dielectric layer **166L** and the gate conductive layer **168L** described with reference to FIG. 1C. The gate dielectric layer **366L** may cover the upper surface and sidewalls of the channel **364** and the upper surface of the

insulating layer **374**. The gate conductive layer **368L** covers the gate dielectric layer **366L**.

[0103] In FIG. **3M**, the gate conductive layer **368L** and the gate dielectric layer **366L** are patterned to remove portions thereof that overlap the electrode contact portion **386C** of the top electrode **386** and expose portions **33** of the top surface of the insulating layer **374**. A gate conductive layer **368L'** and gate dielectric layer **366L'** are formed by removal of the portions that overlap the electrode contact portion **386C**.

[0104] In FIG. **3N**, a third dielectric layer **392L** is formed to cover the gate conductive layer **368L'** and the insulating layer **374**. Formation of the third dielectric layer **392L** is similar in most respects to formation of the oxide layer **160** described with reference to FIG. **1A**.

[0105] In FIG. **3O**, the third dielectric layer **392L** is recessed to form a third dielectric layer **392** and to expose portions of the gate conductive layer **368L'** and the gate dielectric layer **366L'**. As shown, level up an upper surface of the third dielectric layer **392** may be below upper surfaces of the channel **364**, the gate dielectric layer **366L'** and the gate conductive layer **368L'**. This is beneficial so that, following formation of a second insulating layer **393**, at least a portion of the channel **364** may protrude above the second insulating layer **393** for forming an electrode layer **395L** thereon.

[0106] In FIG. **3P**, exposed portions of the gate conductive layer **368L'** and the gate dielectric layer **366L'** above the third dielectric layer **392L** are removed. Then, the second insulating layer **393** is formed. The second insulating layer **393** may be similar in most respects to the insulating layers **174**, **374** described with reference to FIGS. **1G** and **3H**. The second insulating layer **393** may be deposited as a conformal layer that covers the channel **364** and the third dielectric layer **392**. Then, portions of the second insulating layer **393** on the upper surface of the channel **364** may be removed, resulting in the structure depicted in FIG. **3P**. Following removal of the exposed portions, a gate electrode layer **368** and a gate dielectric layer **366** remain.

[0107] In FIG. **3Q**, following formation of the second insulating layer **393**, the electrode layer **395L** may be formed on exposed surfaces of the channel **364** and the second insulating layer **393**. The electrode layer **395L** may be similar in most respects to the first electrode layer **162** described with reference to FIG. **1A**.

[0108] In FIG. **3R**, the electrode layer **395L** is patterned to form an electrode layer **395**, resulting in the structure shown. The electrode layer **395** is in contact with the upper surface of the channel **364** and may be in contact with a portion of sidewalls of the channel **364** that protrude above the second insulating layer **393**. Horizontal portions of the electrode layer **395** are on the upper surface of the second insulating layer **393**.

[0109] In FIGS. **3S** and **3T**, word line, bit line and ground contacts are formed that connect to the gate structure, a source/drain and the fused device of the RRAM device, corresponding to act **2060** of method **2000**. FIG. **3U** is a circuit schematic diagram of the RRAM device **40** in accordance with various embodiments.

[0110] In FIG. **3S**, a fourth dielectric layer **352** is formed over the electrode layer **395** and the second insulating layer **393**. The fourth dielectric layer **352** may be similar in most respects to the first dielectric layer **160** and the dielectric layer **500** described with reference to FIG. **1A** and FIG. **1L**, respectively.

[0111] In FIG. **3T**, contacts **394**, **396**, **398** are formed. The contact **394** lands on the electrode layer **395**, which may be a drain electrode of a transistor **460** including the channel **364**. The contact **396** extends through the fourth dielectric layer **352**, the second insulating layer **393**, the third dielectric layer **392**, the insulating layer **374** and the second dielectric layer **372B** and lands on the electrode contact portion **386C** of the top electrode **386**. The contact **398** extends through the fourth dielectric layer **352**, the second insulating layer **393** and the third dielectric layer **392** and lands on the gate conductive layer **368**. The contacts **394**, **396**, **398** are similar in most respects to the contacts **194**, **196**, **198** described with reference to FIG. **1N**. Following formation of the contacts **394**, **396**, **398**, a second portion of the interconnect structure may be formed on the contacts **394**, **396**, **398**,

corresponding to act 2070 of method 2000.

[0112] FIGS. 4A and 4B are diagrammatic cross-sectional views depicting IC devices 10, 10A that include the RRAM device 40. The IC devices 10, 10A are similar in most respects to the IC devices 10, 10A described with reference to FIGS. 2A, 2B. Instead of including the RRAM device 20, the IC devices 10, 10A include the RRAM device 40. The RRAM device 40 may be positioned in the front side interconnect structure 700 or in the back side interconnect structure 800, or both.

[0113] Embodiments may provide advantages. The RRAM devices 20, 40 have a ground resistor (or memristor) stacked on a source/drain, which reduces area of the RRAM device 20, 40, increasing device density.

[0114] In accordance with at least one embodiment, a device includes: a device layer; and an interconnect structure on the device layer, the interconnect structure including a resistive random access memory (RRAM) device. The RRAM device includes: an electrode layer above the device layer; an oxide semiconductor layer on the electrode layer; a gate structure that wraps around the oxide semiconductor layer; an insulating layer on the gate structure; and a resistor. The resistor includes: a bottom electrode on the oxide semiconductor layer; a dielectric layer on the bottom electrode; and a top electrode on the dielectric layer.

[0115] In accordance with at least one embodiment, a device includes: a device layer; and an interconnect structure on the device layer, the interconnect structure including a resistive random access memory (RRAM) device. The RRAM device includes a resistor that includes: a top electrode above the device layer; a dielectric layer on the top electrode; and a bottom electrode on the dielectric layer. The RRAM device further includes: an oxide semiconductor layer on the bottom electrode; a first insulating layer on the resistor; a gate structure on the first insulating layer, the gate structure wrapping around the oxide semiconductor layer; a second insulating layer on the gate structure; and an electrode layer on the oxide semiconductor layer above the gate structure.

[0116] In accordance with at least one embodiment, a method includes: forming a device layer on a substrate; forming a first portion of an interconnect structure on the device layer; and forming a resistive random access memory (RRAM) device on the first portion, including: forming an oxide semiconductor layer; forming a gate structure that wraps around the oxide semiconductor layer; and forming a resistor, the resistor and the oxide semiconductor layer being stacked in a vertical direction. The method further includes forming a second portion of the interconnect structure on the RRAM device.

[0117] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device, comprising: a device layer; and an interconnect structure on the device layer, the interconnect structure including a resistive random access memory (RRAM) device, the RRAM device including: an electrode layer above the device layer; an oxide semiconductor layer on the electrode layer; a gate structure that wraps around the oxide semiconductor layer; an insulating layer on the gate structure; and a resistor including: a bottom electrode on the oxide semiconductor layer; a dielectric layer on the bottom electrode; and a top electrode on the dielectric layer.
2. The device of claim 1, further comprising: a first contact that extends through the insulating layer and lands on the electrode layer; a second contact that lands on the top electrode; and a third

contact that extends through the insulating layer and lands on the gate structure.

3. The device of claim 1, wherein the RRAM device further includes: at least one spacer on sidewalls of the top electrode.

4. The device of claim 3, wherein the at least one spacer includes: an upper spacer on the sidewalls of the top electrode; and a lower spacer on sidewalls of the top electrode, the dielectric layer and the bottom electrode and on an upper surface of the insulating layer.

5. The device of claim 1, wherein the RRAM device includes: an oxygen reservoir layer between the dielectric layer and the top electrode.

6. The device of claim 1, wherein the RRAM device has a sharp-tipped profile in cross-sectional side view.

7. The device of claim 1, wherein the RRAM device has an M-shaped profile in cross-sectional side view.

8. A device, comprising: a device layer; and an interconnect structure on the device layer, the interconnect structure including a resistive random access memory (RRAM) device, the RRAM device including: a resistor including: a top electrode above the device layer; a dielectric layer on the top electrode; and a bottom electrode on the dielectric layer; an oxide semiconductor layer on the bottom electrode; a first insulating layer on the resistor; a gate structure on the first insulating layer, the gate structure wrapping around the oxide semiconductor layer; a second insulating layer on the gate structure; and an electrode layer on the oxide semiconductor layer above the gate structure.

9. The device of claim 8, further comprising: a first contact that extends through the first and second insulating layers and lands on the top electrode; a second contact that lands on the electrode layer; and a third contact that extends through the second insulating layer and lands on the gate structure.

10. The device of claim 8, wherein: the interconnect structure includes a second dielectric layer and an opening in the second dielectric layer; and the top electrode, bottom electrode and dielectric layer of the RRAM device and the oxide semiconductor layer extend into the opening.

11. The device of claim 10, wherein a second portion of the oxide semiconductor layer that is above the opening has width that exceeds that of a first portion of oxide semiconductor layer that is in the opening.

12. The device of claim 8, wherein the RRAM device includes an oxygen reservoir layer between the dielectric layer and the top electrode.

13. The device of claim 8, further comprising: a second interconnect structure on a front side of the device layer; wherein the interconnect structure is a back side interconnect structure on a back side of the device layer opposite the front side.

14. The device of claim 13, wherein: the device layer includes a nanostructure transistor including a stack of nanostructure channels and a source/drain; and the interconnect structure includes: a first power rail that connects to the source/drain; and a second power rail on the first power rail, the second power rail connecting to the RRAM device, the second power rail having dimension that exceeds that of the first power rail.

15. A method, comprising: forming a device layer on a substrate; forming a first portion of an interconnect structure on the device layer; forming a resistive random access memory (RRAM) device on the first portion, including: forming an oxide semiconductor layer; forming a gate structure that wraps around the oxide semiconductor layer; and forming a resistor, the resistor and the oxide semiconductor layer being stacked in a vertical direction; and forming a second portion of the interconnect structure on the RRAM device.

16. The method of claim 15, wherein the forming a resistor is after the forming an oxide semiconductor layer.

17. The method of claim 15, wherein the forming a resistor includes: forming a bottom electrode on the oxide semiconductor layer; forming a switching layer on the bottom electrode; and forming a

top electrode on the switching layer.

18. The method of claim 17, wherein the forming a resistor includes forming the bottom electrode, the switching layer and the top electrode having a sharp-tipped profile in cross-sectional side view.

19. The method of claim 17, wherein the forming a resistor includes forming the bottom electrode, the switching layer and the top electrode having an M-shaped profile in cross-sectional side view.

20. The method of claim 15, wherein the forming a first portion of an interconnect structure is forming the first portion of a back side interconnect structure, the method further including: forming a front side interconnect structure prior to the forming a first portion of an interconnect structure.
